



THE 7-STAGE SENSE-TO-ACTUATION LATENCY WATERFALL

Where Does the Time Go? Deconstructing the Physical, Memory, Neural, and Silicon Bottlenecks



STAGE 1

Transduce

Photodiode Well Exposure
& IMU Deflection

P_{50} : 8 ms

P_{99} : 15 ms



STAGE 2

DMA Ingest

MIPI CSI-2 Bus Transfer
into Shared DRAM

P_{50} : 3 ms

P_{99} : 8 ms



STAGE 3

Perception

ViT Encoders & DINOv2
3D Spatial Tokens

P_{50} : 12 ms

P_{99} : 25 ms



STAGE 4

Policy VLA

Diffusion ACT / VLA
Action Chunk Pass

P_{50} : 22 ms

P_{99} : 60 ms



STAGE 5

Inter-IPC

Shared SRAM Mailbox
RPMSG MPU-to-MCU

P_{50} : 0.8 ms

P_{99} : 2.0 ms



STAGE 6

Enforce

1 kHz CBF Filter &
Stopping Distance

P_{50} : 0.4 ms

P_{99} : 1.0 ms



STAGE 7

Actuator

Motor Stator L/R Coil
Current Rise to Torque

P_{50} : 6 ms

P_{99} : 15 ms



Nominal Path ($P_{50} = 52.2$ ms):



Tail Path ($P_{99} = 126.0$ ms):

Safe Closed-Loop Margin (52.2 ms $\ll \tau_{\text{world}}$)

CRITICAL TIMING VIOLATION (126.0 ms $> \tau_{\text{world}}$)

World Deadline $\tau_{\text{world}} = 100$ ms